

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

**Artificial Intelligence [ELE-4061]**

**Marks: 50**

**Duration: 180 mins.**

### PART-A

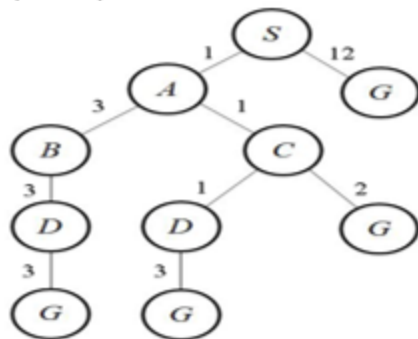
**Answer all the questions.**

- 1) What are the components of AI? Give one line description for each of them. (3)
  - A)
  - B) What is PEAS descriptor? Explain PEAS for automated Taxi Driver Agent. (3)
  - C) Explain the artificial agent terminology with a suitable example. (4)
- 2) Give a practical example for goal-based agent and explain with block diagram. (3)
  - A)
  - B) Develop the state space representation for missionaries and carnivals. Problem definition: Three missionaries and three cannibals are on one side of the river. They have a boat that can hold a maximum of two people. Find a way to get everyone on the other side without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place. Give solution steps. That is number of missionaries should not be less than number of cannibals on any of the sides of the river. (3)



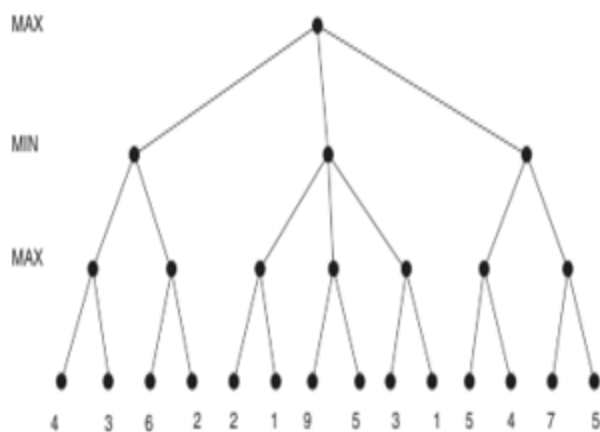
- C) (4)

Apply uniform cost search to find the path from S to G. Clearly show all the iterations for getting the optimal path.



- 3) Develop the solution tree using alpha-beta pruning. Clearly show all the steps.

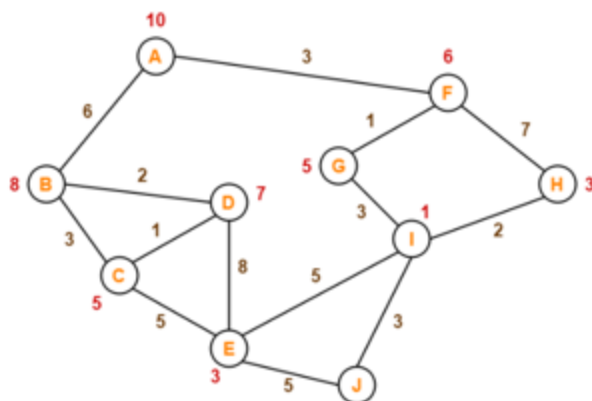
A)



(3)

B)

Apply A\* algorithm to find the most cost-effective path from start state A to final state J. The numbers written on the edges represent the distance between the nodes. The numbers written on the node represent the heuristic value. Clearly show all the steps.



(3)

C)

Solve using Crypto arithmetic technique: WINTER + SUMMER = AUTUMN

(4)

- 4) Explain Quantifiers in first order logic. Also, list the properties of Quantifiers.

(3)

A)

B)

Convert the following WFF to CNF:

$$(P \longrightarrow Q) \longrightarrow R$$

(3)

C)

Consider the following axioms.

1. If it is sunny and warm day, you will enjoy
2. If it is raining, you will get wet
3. It is warm day
4. It is raining
5. It is sunny
6. Goal: you will enjoy.

(4)

Prove using Resolution.

5)

Two students, Anil and Arun, live in the same city and come to college by two different means. Anil comes by train and Arun comes by bus. The railway workers go on strike sometimes, which affects both and hence they come to college late.

A)

1. Draw the DAG
2. Determine the probability of both being late for college when there is a railway strike.
3. Determine the probability of both being late for college when there is no strike.

| Train Strike | Probability |
|--------------|-------------|
| Y            | 0.1         |
| N            | 0.9         |

(3)

|                |   | Train Strike |     |
|----------------|---|--------------|-----|
|                |   | Y            | N   |
| Sachin<br>Late | Y | 0.9          | 0.1 |
|                | N | 0.1          | 0.9 |

|               |   | Train Strike |     |
|---------------|---|--------------|-----|
|               |   | Y            | N   |
| Rahul<br>Late | Y | 0.2          | 0.1 |
|               | N | 0.8          | 0.9 |

B)

(3)

Consider the learning task represented by the training examples as shown below. Apply Naïve Bayes classifier, predict and assign target value to 'Play Tennis', if the present condition is sunny, cool, normal humidity and weak wind.

| Day | Outlook  | Temperature | Humidity | Wind   | Play Tennis |
|-----|----------|-------------|----------|--------|-------------|
| D1  | Sunny    | Hot         | High     | Weak   | No          |
| D2  | Sunny    | Hot         | High     | Strong | No          |
| D3  | Overcast | Hot         | High     | Weak   | Yes         |
| D4  | Rain     | Cool        | High     | Weak   | Yes         |
| D5  | Rain     | Cool        | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool        | Normal   | Strong | No          |
| D7  | Overcast | Mild        | Normal   | Strong | Yes         |
| D8  | Sunny    | Cool        | High     | Weak   | No          |
| D9  | Sunny    | Mild        | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild        | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild        | Normal   | Strong | Yes         |
| D12 | Overcast | Hot         | High     | Strong | Yes         |
| D13 | Overcast | Hot         | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild        | High     | Strong | No          |

- C) Determine which species does a flower with sepal length 5.2, sepal width 3.1 belong to using KNN algorithm. Assume K=3

| Sepal Length | Sepal Width | Species    |
|--------------|-------------|------------|
| 5.3          | 3.7         | Setosa     |
| 5.1          | 3.8         | Setosa     |
| 7.2          | 3.0         | Virginica  |
| 5.4          | 3.4         | Setosa     |
| 5.1          | 3.3         | Setosa     |
| 5.4          | 3.9         | Setosa     |
| 7.4          | 2.8         | Virginica  |
| 6.1          | 2.8         | Versicolor |
| 7.3          | 2.9         | Virginica  |
| 6.0          | 2.7         | Versicolor |
| 5.8          | 2.8         | Virginica  |
| 6.3          | 2.3         | Versicolor |
| 5.1          | 2.5         | Versicolor |
| 6.3          | 2.5         | Versicolor |
| 5.5          | 2.4         | Versicolor |

(4)

-----End-----